TRUNCATIONS OF THE RING OF NUMBER-THEORETIC FUNCTIONS, REVISITED

JAN SNELLMAN

Matematiska Institutionen, Linköpings Universitet, 581 83 Linköping, Sweden
jan.snellman@liu.se

Abstract. Let K be a field containing $\mathbb{Q}$, let Γ be the ring of all functions $\mathbb{N}^+ \to K$ under Dirichlet convolution, and let Γ_n be its truncation to functions supported on $[1, n]$. In [9] it was shown that Γ_n is a polynomial ring modulo a monomial ideal I_n which is stable after reversing the order of the variables, and the Poincaré–Betti series of Γ_n was computed in terms of the numbers $C_{n,v}$ of minimal generators of I_n of least support v.

We prove that $C_{n,v} = \Phi(n, p_v)$, Legendre's sifting function: the number of integers in $[1, n]$ free of prime factors $\leq p_v$. This identifies an invariant of a minimal free resolution with a classical object of sieve theory. As consequences we obtain: a proof of Conjecture 4.6 of [9], which was left open there; the average order $C_n \sim \pi(n)^2/2$ of the total number of minimal generators, showing that the lower bound $C_n \geq \binom{\pi(n)+1}{2}$ of [9] is asymptotically sharp, together with the exact order $C_n - \binom{\pi(n)+1}{2} \sim \frac{16}{3} n^{3/2}/\log^3 n$ of the error; and the identification of C_n with the OEIS sequence A182843. We also record errata for [9]: one stated result is false, and two proofs are incomplete. Corrected statements and complete proofs are given.

1. Introduction

Cashwell and Everett [2] studied the ring

$$\Gamma = \{ f \colon \mathbb{N}^+ \to K \} \tag{1}$$

of *number-theoretic functions*, with pointwise addition and the Dirichlet convolution

$$fg(n) = \sum_{ab=n} f(a)g(b); \tag{2}$$

here K is a field containing $\mathbb{Q}$. Reading the fundamental theorem of arithmetic as an isomorphism of monoids

$$(\mathbb{N}^+, \cdot) \;\cong\; \coprod_p (\mathbb{N}, +) \tag{3}$$

identifies Γ with the large power series ring $K[[x_1, x_2, \dots]]$ on countably many variables, x_i corresponding to the i-th prime p_i; Cashwell and Everett proved that this ring is a unique factorisation domain.

In [9] the *truncations*

$$\Gamma_n = \{ f \in \Gamma : f(m) = 0 \text{ for all } m > n \} \tag{4}$$

were studied. Writing $r = \pi(n)$ and giving the monomial $x_1^{a_1} \cdots x_r^{a_r}$ the *weight* $\prod_i p_i^{a_i}$, so that monomials correspond bijectively to positive integers, one has

$$\Gamma_n \cong S/I_n, \qquad S = K[x_1, \ldots, x_r], \tag{5}$$

with I_n generated by all monomials of weight $> n$. The ideal I_n is strongly stable with respect to the *reversed* order of the variables, because $w(mx_j/x_i) = w(m)p_j/p_i > w(m)$ whenever $j \geq i$, that is

$$w(m\,x_j/x_i) = w(m)\,\frac{p_j}{p_i} > w(m); \tag{6}$$

this single inequality puts the Eliahou–Kervaire resolution [3] and the Golod property [1, 7, 5, 6] within reach, and yields a rational Poincaré–Betti series for Γ_n expressed in terms of the numbers

$$C_{n,v} = \#\{\, m \in G(I_n) : \min(m) = v \,\}, \qquad C_n = \#G(I_n), \tag{7}$$

where $G(I_n)$ is the minimal monomial generating set and $\min(m)$ is the least index occurring in m.

The purpose of this note is threefold.

First, *errata*. Theorem 3.5(3) of [9] is false, and the proofs of Theorem 3.4(4) and Theorem 3.4(5) there are incomplete. Section 4 records this and supplies corrected statements and complete proofs. The false statement is used nowhere else in [9], so nothing else in that paper is affected.

Second, and principally, a *sieve-theoretic identification*. Recall Legendre's sifting function

$$\Phi(x, y) = \#\{\, m \leq x : p \mid m \implies p > y \,\}, \tag{8}$$

the number of integers up to x free of prime factors $\leq y$. Our main result, Theorem 4, is that

$$\boxed{C_{n,v} = \Phi(n, p_v).} \tag{9}$$

Thus the generator counts governing a minimal free resolution over Γ_n are literally values of the classical sifting function, and the numerical tables of [9] are tables of Φ. Theorem 3.2 of [9] describes $C_{n,v}$ as the cardinality of a *different* set — the p_v-rough numbers in the interval $(n/p_v, n]$ — and Theorem 4 may be read as saying that this set is equinumerous with the much more familiar one in (8).

Third, the *consequences*. Section 5 proves Conjecture 4.6 of [9], which asserted a precise shape for the reduced Poincaré–Betti series and was left open there with numerical evidence for $n \leq 25$. Section 6 determines the average order of C_n, showing that

$$C_n \sim \tfrac{1}{2}\pi(n)^2 \sim \frac{n^2}{2\log^2 n},$$

and hence that the lower bound $C_n \geq \binom{\pi(n)+1}{2}$ of Theorem 3.4(3) of [9] is asymptotically an equality. Section 9 collects what remains open.

Remark 1. The Golod criterion of [1, 7] requires the ideal to be contained in $(x_1, \ldots, x_r)^2$, and I_n is: a degree-one generator would be some x_i with $p_i > n$, and no such variable occurs in $S = K[x_1, \ldots, x_r]$ since $r = \pi(n)$. Equivalently, this is Theorem 3.5(1) of [9]. We record it because it is exactly the sort of hypothesis that goes unchecked.

Remark 2. All numerical claims below were verified by computer, along a route independent of the formulas being tested: the minimal generators of I_n were enumerated directly and their least supports tabulated, rather than computed from any of the counting formulas. The code and its raw output are available in the repository [29].

2. Notation and the counting theorem

Throughout, $p_1 = 2 < p_2 = 3 < \cdots$ are the primes, $\pi(n)$ is the number of primes $\leq n$, and $\operatorname{lpf}(m)$ denotes the least prime factor of $m > 1$. We abbreviate $r = \pi(n)$ when n is fixed, this being the number of variables of S. ([9] writes $r(n)$ throughout for what we write $\pi(n)$; we have changed the notation because the results below sit alongside the sieve-theoretic literature, where π is universal. Quotations from [9] below keep its own notation.) For a monomial $m = x_1^{a_1} \cdots x_r^{a_r}$ we write $\min(m)$ and $\max(m)$ for the least and greatest i with $a_i > 0$, and $|m| = \sum_i a_i$. For an integer $m > 1$, $\Omega(m)$ denotes the number of prime factors of m counted with multiplicity, so that $\Omega(p_1^{a_1} \cdots p_r^{a_r}) = a_1 + \cdots + a_r$, with $\Omega(1) = 0$; following [9] we also write λ for Ω. The companion function counting prime factors *without* multiplicity, usually written ω, is not used in this note. Where the symbol ω does appear — in Remark 27 and in Section 8 — it always denotes Buchstab's function, an unrelated object.

We shall use the following description, which is Theorem 3.2 of [9].

Theorem 3 ([9], Thm. 3.2)**.** *For $1 \leq v \leq \pi(n)$,*

$$C_{n,v} = \#\{\, x \in \mathbb{N} : n/p_v < x \leq n \text{ and } p \mid x \implies p \geq p_v \,\}.$$

It will be convenient to name the counting function appearing implicitly there. For a prime p and real $y \geq 0$ put

$$R_p(y) = \#\{\, m \leq y : q \mid m \implies q \geq p \,\}, \tag{10}$$

the number of p-rough integers up to y; the integer 1 is counted, vacuously. Theorem 3 then reads

$$C_{n,v} = R_{p_v}(n) - R_{p_v}(n/p_v). \tag{11}$$

Note that

$$R_p(y) = \Phi(y, p) + \#\{\, m \leq y : \operatorname{lpf}(m) = p \,\}, \tag{12}$$

so R_p and Φ differ precisely by the multiples of p that are p-rough.

3. The main theorem

Theorem 4. *For all $n \geq 2$ and $1 \leq v \leq \pi(n)$,*

$$C_{n,v} = \Phi(n, p_v),$$

the number of integers in $[1, n]$ having no prime factor $\leq p_v$.

Proof. Write $q = p_v$. Every q-rough integer m factors uniquely as $m = q^a m'$ with $a \geq 0$ and m' free of prime factors $\leq q$; conversely every such pair (a, m') gives a q-rough integer. Counting those with $m \leq y$ according to a gives

$$R_P(y) = \sum_{a \geq 0} \Phi(y/q^a,\ q)\,, \tag{13}$$

a finite sum, since the terms vanish once $q^a > y$. Substituting (13) into (11), the two sums telescope:

$$\begin{aligned} C_{n,v} &= R_P(n) - R_P(n/q) \\ &= \sum_{a\ge 0} \Phi\left(\frac{n}{q^a}, q\right) - \sum_{a\ge 0} \Phi\left(\frac{n}{q^{a+1}}, q\right) \\ &= \Phi(n,q). \end{aligned}$$

□

Example 5. Take $n = 25$, so $r = 9$. The integers in $[1, 25]$ with no prime factor ≤ 2 are the thirteen odd numbers, so $C_{25,1} = 13$; those with no prime factor ≤ 3 are $1, 5, 7, 11, 13, 17, 19, 23, 25$, so $C_{25,2} = 9$; those with no prime factor ≤ 5 are $1, 7, 11, 13, 17, 19, 23$, so $C_{25,3} = 7$. These agree with Figure 1 of [9]. By contrast the set of Theorem 3 for $v = 2$ is $\{9, 11, 13, 15, 17, 19, 21, 23, 25\}$ — a different set of the same size.

Splitting $[1, n]$ into 1, the primes and the composites gives at once the following reformulation, which is the form we shall use.

Corollary 6. *For* $1 \le v \le \pi(n)$,

$$C_{n,v} = \big(\pi(n) - v + 1\big) + E(n, v), \tag{14}$$

where

$$E(n, v) = \#\{\, x \le n : x \text{ composite},\ p \mid x \implies p > p_v \,\}. \tag{15}$$

In particular

$$C_{n,v} \ \ge\ \pi(n) - v + 1,$$

with equality if and only if $p_{v+1}^2 > n$.

Proof. The integers counted by $\Phi(n, p_v)$ are 1; the primes q with $p_v < q \le n$, of which there are $\pi(n) - v$; and the composites counted by $E(n, v)$. For the last sentence, $E(n, v) = 0$ exactly when there is no composite $\le n$ all of whose prime factors exceed p_v, and the least such composite is p_{v+1}^2. □

Corollary 6 contains Theorem 3.4(2) of [9] (put $v = 1 + \pi(n) - u$) and sharpens it to an exact criterion for equality. Summing over v gives a closed form for the total.

Corollary 7. $C_n = \binom{\pi(n)+1}{2} + \sum\limits_{\substack{x \le n \\ x \text{ composite}}} \big(\pi(\operatorname{lpf} x) - 1\big).$

Proof. Sum Corollary 6 over $v = 1, \ldots, r$. The first terms contribute

$$\sum_{v=1}^{r} (r - v + 1) = \sum_{j=1}^{r} j = \binom{r+1}{2}.$$

For the second, exchange the order of summation: a composite $x \le n$ is counted by $E(n, v)$ for exactly those $v \ge 1$ with $p_v < \operatorname{lpf} x$, and there are $\pi(\operatorname{lpf} x) - 1$ of those. □

Remark 8. Corollary 7 proves the increment formula

$$C_n - C_{n-1} = \begin{cases} \pi(n) & \text{if } n \text{ is prime}, \\ \pi(\operatorname{lpf} n) - 1 & \text{if } n \text{ is composite}, \end{cases}$$

recorded empirically by Costello for the OEIS sequence A182843; see Section 6. The two cases have different sources: when n is prime the whole increment comes from $\binom{\pi(n)+1}{2}$, which grows by $\pi(n)$, and the sum over composites is unchanged; when n is composite the binomial is unchanged and the sum gains the single term $\pi(\operatorname{lpf} n) - 1$. They can be written as one expression, $[\,n \text{ prime}\,] + \big(\pi(\operatorname{lpf} n) - 1\big)$ with an Iverson bracket, because $\operatorname{lpf} n = n$ for n prime.

4. Errata for [9]

4.1. A false statement. Theorem 3.5(3) of [9] asserts

$$\binom{\pi(n)}{2} = \#\{\, m \in \mathbb{N}^+ : m \le n,\ \lambda(m) = 2 \,\}, \tag{16}$$

and the proof there reads, in full, "The first and the last assertions are obvious." Statement (16) is false. Its left-hand side counts *pairs of distinct primes* $\le n$, its right-hand side counts *products of two primes that are* $\le n$; the two agree only for $n = 2$ and $n = 4$. At $n = 10$ the left side is $\binom{4}{2} = 6$ while the right side counts $\{4, 6, 9, 10\}$, so is 4. Asymptotically the left side is $\sim n^2/(2\log^2 n)$ and the right side $\sim n \log\log n/\log n$.

The intended statement was presumably the following, which is what the proof of Theorem 3.4(2)–(3) of [9] actually establishes.

Proposition 9. *For every $n \ge 2$,*

$$\#\{\, m \in G(I_n) : |\operatorname{supp}(m)| = 2 \,\} \ \ge\ \binom{\pi(n)}{2},$$

with equality exactly for $2 \le n \le 8$.

Proof. Let $1 \le i < j \le r$. Choose $b \ge 1$ with $p_i^{b-1} \le n < p_i^b$. Then

$$w(x_i^{b-1} x_j) = p_i^{b-1} p_j > p_i^{b-1} p_i = p_i^b > n,$$

so $x_i^{b-1} x_j \in I_n$ and is therefore divisible by some $m \in G(I_n)$, necessarily with $\operatorname{supp}(m) \subseteq \{i, j\}$. Now $x_i^{b-1} \notin I_n$ because $p_i^{b-1} \le n$, so m is not a power of x_i; and $x_j \notin I_n$ because $p_j \le n$, so m is not a power of x_j. Hence $\operatorname{supp}(m) = \{i, j\}$. Distinct pairs give distinct generators, whence the inequality.

For the equality range, count the generators supported on $\{1, 2\}$ exactly. Let $b^* = \max\{\, b \ge 1 : 3^b \le n \,\}$, and for $1 \le b \le b^*$ put $a(b) = \min\{\, a \ge 1 : 2^a 3^b > n \,\}$. Then $x_1^{a(b)} x_2^b$ is a minimal generator: its weight exceeds n by definition; $2^{a(b)-1} 3^b \le n$ by minimality of $a(b)$ when $a(b) \ge 2$, and because $3^b \le n$ when $a(b) = 1$; and

$$2^{a(b)} 3^{\,b-1} = \tfrac{1}{3} \cdot 2^{a(b)} 3^b \le \tfrac{1}{3} \cdot 2 \cdot 2^{a(b)-1} 3^b \le \tfrac{2}{3} n \le n.$$

Conversely every generator supported on $\{1, 2\}$ arises this way, since $2^{a-1} 3^b \le n$ forces $b \le b^*$, and a is then determined. So there are exactly b^* of them.

Hence the inequality is strict as soon as $b^* \ge 2$, that is, as soon as $n \ge 9$; and for $2 \le n \le 8$ equality is a finite check. ∎

4.2. **Two incomplete proofs.** *Theorem 3.4(4) of* [9] asserts that for each v one has $C_{n,1+\pi(n)-v} = v$ for all sufficiently large n. The proof given there reads "the only integers $x \le n$ with all prime factors $\ge 1 + r(n) - v$ are $p_{1+r(n)-v}, \dots, p_{r(n)}$... and they are all $> n/p_v$". Two subscripts are wrong: the condition on prime factors should read $\ge p_{1+r(n)-v}$ (a prime, not an index), and the final bound should read $> n/p_{1+r(n)-v}$. With Corollary 6 the statement becomes immediate, and effective.

Proposition 10. *Fix $v \ge 1$. Then $C_{n,1+\pi(n)-v} = v$ for every $n > 4^{v-1}$.*

Proof. Put $u = 1+\pi(n)-v$. By Corollary 6, $C_{n,u} = v$ if and only if $p_{u+1}^2 > n$, where $p_{u+1} = p_{\pi(n)-v+2}$. By Bertrand's postulate there is a prime in $(y, 2y)$ for every $y > 1$; applied repeatedly this gives $p_{\pi(n)-j+1} > n/2^j$ for $j \ge 1$, while for $j = 0$ the same inequality reads $p_{\pi(n)+1} > n$, true by definition of π. So $p_{\pi(n)-v+2} > n/2^{v-1}$ and hence $p_{u+1}^2 > n^2/4^{v-1}$, which exceeds n as soon as $n > 4^{v-1}$. □

Theorem 3.4(5) of [9] asserts that $C_{n,v} = C_{n-1,v}$ for all v when n is even. The proof there establishes only one of the two inclusions needed. It shows that an x counted for n is counted for $n-1$; the converse, that no x is counted for $n-1$ but not for n, is not addressed. The statement is nonetheless true, and with Theorem 4 the whole matter is a triviality.

Proposition 11. *If $n \ge 4$ is even then $C_{n,v} = C_{n-1,v}$ for $1 \le v \le \pi(n)$; for $v > \pi(n)$ both sides vanish.*

Proof. By Theorem 4, $C_{n,v} = \Phi(n, p_v)$ and $C_{n-1,v} = \Phi(n-1, p_v)$. These differ by 1 or 0 according as n is or is not counted by $\Phi(\cdot, p_v)$; and n is even, so it is divisible by $p_1 = 2 \le p_v$ and is never counted. □

Remark 12. For completeness we note that the argument omitted in [9] can also be supplied directly, *for* $v > 1$: an integer lying in $((n-1)/p_v,\, n/p_v]$ is unique if it exists, since that interval has length $1/p_v < 1$, and it must satisfy $p_v x = n$; as n is even and p_v odd for $v > 1$, such an x is even and so is excluded by the condition on prime factors. At $v = 1$ this fails: the roughness condition is then vacuous, and the two sets of Theorem 3 genuinely differ, neither containing the other, their cardinalities agreeing only after a compensating gain at the top end. That case is [9]'s own item (6), $C_{n,1} = \lceil n/2 \rceil$, and needs no repair.

4.3. **Smaller items.** For the record: in equation (15) of [9] the range should be $1 \le j \le r$, not $1 \le j \le n$, since j indexes the variables $x_1, \dots, x_r$; in Theorem 3.2 and equation (19) the solution vector is written $(b_1, \dots, b_r) \in \mathbb{N}^r$ although only $b_v, \dots, b_r$ are constrained, so that as literally stated the solution set is infinite for $v > 1$ — the proof supplies the missing condition $b_j = 0$ for $j < v$ immediately, and the corresponding statement in Theorem 1.2(III) is correct as printed; the caption of Figure 2 names the quantities $C_{n,i,g}$ where the text defines $C_{n,v,d}$; and Theorem 4.3 is attributed to "Herzog–Aramova" in the text but "Aramova–Herzog" in the abstract and bibliography. Finally, we take the opportunity to note a small looseness of attribution. Theorem 4.3 of [9] is stated for *stable* ideals and credited jointly to [1] and [7]. The result of [7] (Corollary 1.2 there) assumes the ideal to be 0-Borel fixed, that is *strongly* stable, and it is in that form that the joint attribution is standard; see [25, Thm. 6.16(6)]. The version for merely stable ideals is obtained in [1], §2, where the Koszul cycles exhibited for a stable ideal are observed to have pairwise vanishing products. None of this affects [9], whose Proposition 2.6 proves I_n strongly stable, so that both hypotheses hold; we note it only because the distinction matters if the argument is reused. An alternative route to the same conclusion runs through componentwise linearity [22], since stable ideals have linear quotients.

5. Conjecture 4.6

By Corollary 4.4 of [9], the Poincaré–Betti series of the residue field over $A_n = \Gamma_n$ is

$$P_K^{A_n}(t) = \frac{(1+t)^r}{D_n(t)}, \qquad D_n(t) = 1 - t^2 \sum_{j=1}^{r} (1+t)^{r-j} C_{n,j}, \tag{17}$$

with $r = \pi(n)$. Conjecture 4.6 of [9] asserted that this fraction reduces to

$$P_K^{A_n}(t) = -\frac{(1+t)^{\ell_1(n)}}{q_n(t)}, \qquad q_n(t) = \sum_{i=0}^{\ell_2(n)} h_i(n) t^i, \tag{18}$$

where

(1) $q_n(-1) \neq 0$;
(2) $\ell_1(n) = \#\{\, p \text{ odd prime} : p^2 \leq n \,\}$;
(3) $\ell_2(n) = \ell_1(n) + 1$;
(4) $h_0(n) = -1$;
(5) $h_1(n) = \pi(n) - \ell_1(n)$;
(6) $h_{\ell_2(n)}(n) = C_{n,1} = \lceil n/2 \rceil$.

We prove all six.

The point of the next lemma is that the whole conjecture is controlled by a single quantity: the order of vanishing of D_n at $t = -1$.

Lemma 13. *Put $u = 1 + t$ and $c_k = C_{n,r-k}$ for $0 \leq k \leq r-1$, with $c_k = 0$ otherwise. Then the coefficient of u^0 in D_n is $1 - c_0$, and for $k \geq 1$ the coefficient of u^k is $-(c_k - 2c_{k-1} + c_{k-2})$.*

Proof. Reindexing (17) by $k = r - j$ gives $D_n = 1 - (u-1)^2 T(u)$ with $T(u) = \sum_{k=0}^{r-1} c_k u^k$; expand $(u-1)^2 = u^2 - 2u + 1$. □

Lemma 14. *$c_0 = C_{n,\pi(n)} = 1$ for every $n \geq 2$. Consequently $D_n(-1) = 0$.*

Proof. $\Phi(n, p_r)$ counts the integers in $[1, n]$ with no prime factor $\leq p_r$; as p_r is the largest prime $\leq n$, the only such integer is 1. Now apply Theorem 4 and Lemma 13. □

Proposition 15. *The order of vanishing of $D_n(t)$ at $t = -1$ equals $\pi(n) - \ell_1(n)$.*

Proof. By Lemmas 13 and 14 the order is the least $k \geq 1$ with $c_k - 2c_{k-1} + c_{k-2} \neq 0$. Since $c_{-1} = 0$ and $c_0 = 1$, an induction shows that the second differences vanish for all $k \le K$ precisely when $c_k = k+1$ for all $k \le K$. So the order is the least $k \ge 1$ with

$$c_k \neq k+1, \qquad \text{that is,} \qquad C_{n,r-k} \neq r-(r-k)+1.$$

By Corollary 6,

$$C_{n,v} \neq r-v+1 \iff p_{v+1}^2 \le n \iff p_{v+1} \le \sqrt{n}.$$

Suppose first that some admissible v has this property; since $p_2^2 = 9$, that requires $n \ge 9$. The largest such v is then

$$v = \pi\big(\lfloor\sqrt{n}\rfloor\big) - 1 = \#\{\, p \text{ odd} : p^2 \le n \,\} = \ell_1(n) \qquad (n \ge 9),$$

the two counts differing only by the prime 2. Hence the least admissible k is $r - \ell_1(n)$. (The identity between the two counts in fact holds for every $n \ge 4$; what needs $n \ge 9$ is that the set of such v is non-empty.)

If no such v exists — equivalently $\ell_1(n) = 0$, equivalently $n \le 8$ — then $c_k = k+1$ for all $k \le r-1$, and the first non-vanishing coefficient is that of u^r, namely

$$-\big(0 - 2r + (r-1)\big) = r+1 \neq 0;$$

the formula $r - \ell_1(n) = r$ still holds. □

Theorem 16. *Let $n \ge 2$. Conjecture 4.6 of* [9] *holds: with $\ell_1(n)$ and $\ell_2(n)$ as above,* (18) *holds together with clauses (1)–(6). (At $n = 1$ the statement degenerates: $r = 0$, $D_1 = 1$, and clause (3) fails,* $\deg q_1 = 0 \neq 1 = \ell_2(1)$.*)*

Proof. By Proposition 15 we may write $D_n(t) = (1+t)^{r-\ell_1} d(t)$ with $d(-1) \neq 0$; set $q_n = -d$. Then (17) gives

$$P_K^{A_n} = \frac{(1+t)^r}{(1+t)^{r-\ell_1}(-q_n)} = -\frac{(1+t)^{\ell_1}}{q_n},$$

which is (18) together with clause (2). Clause (1) is $d(-1) \neq 0$.

For the remaining clauses, note first that $D_n(0) = 1$ and $D_n'(0) = 0$, since $D_n - 1 = -t^2(\cdots)$ vanishes to order 2 at $t = 0$; and that $\deg D_n = r+1$ with leading coefficient $-C_{n,1}$, the top-degree term of (17) arising only from $j = 1$.

(3) We have

$$\deg q_n = \deg D_n - (r - \ell_1) = (r+1) - (r - \ell_1) = \ell_1 + 1 = \ell_2.$$

(4) $h_0 = q_n(0) = -D_n(0)/1^{r-\ell_1} = -1$.

(5) Differentiating $q_n = -D_n(1+t)^{-(r-\ell_1)}$ at $t = 0$ gives

$$h_1 = q_n'(0) = -D_n'(0) + (r-\ell_1)D_n(0) = r - \ell_1.$$

(6) The leading coefficient of q_n is minus that of D_n, namely $C_{n,1}$, which is $\Phi(n,2) = \lceil n/2 \rceil$ by Theorem 4. □

Remark 17. Clause (2) — the identification of the numerator exponent — was the whole difficulty, and Proposition 15 shows why: it is the assertion that the sequence $v \mapsto C_{n,v}$ is exactly linear, of slope -1, on the range $v > \ell_1(n)$, and departs from linearity at $v = \ell_1(n)$. Corollary 6 makes both halves of that assertion visible at once, because it measures the departure from linearity by $E(n,v)$, a count of composites all of whose prime factors exceed p_v, which is positive exactly when $p_{v+1}^2 \le n$. Figure 1 shows the effect.

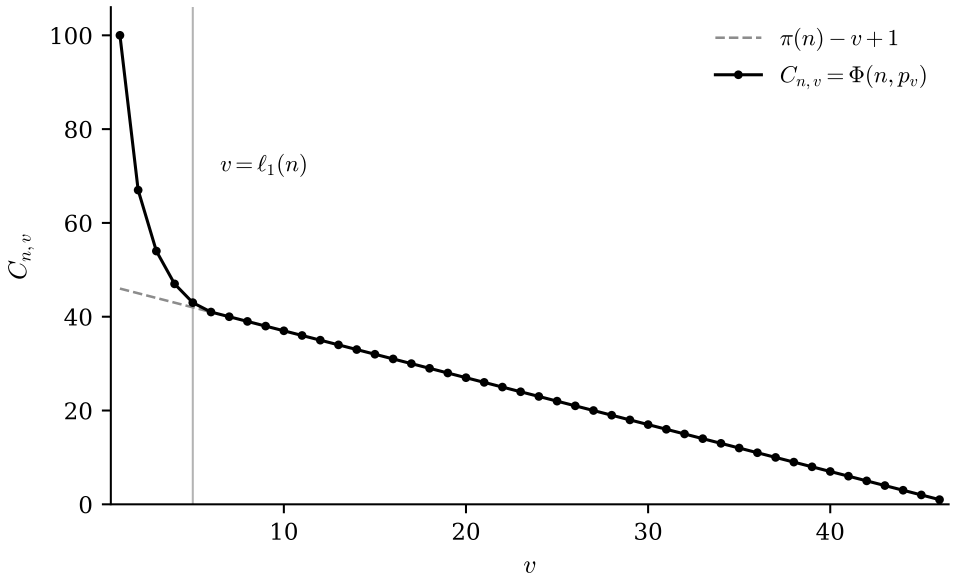

FIGURE 1. The profile $v \mapsto C_{n,v} = \Phi(n, p_v)$ for $n = 200$, against the line $\pi(n) - v + 1$ of Corollary 6. The vertical distance between them is $E(n, v)$, the number of composites $\leq n$ all of whose prime factors exceed p_v; here $\pi(n) = 46$ and $\ell_1(n) = 5$, and the two coincide for every $v > 5$. That the profile is exactly linear of slope -1 beyond $v = \ell_1(n)$, and only there, is what Proposition 15 extracts as the order of vanishing of D_n at $t = -1$.

6. THE AVERAGE ORDER OF C_n

Theorem 3.4(3) of [9] gives the lower bound $C_n \geq \binom{\pi(n)+1}{2}$, proved there by exhibiting one minimal generator for each support of size 1 or 2. We show that essentially nothing else contributes in the limit.

Theorem 18. *For $n \geq 4$, $0 \leq C_n - \binom{\pi(n)+1}{2} \leq n\big(\pi(\sqrt{n}) - 1\big) = O\Big(n^{3/2}/\log n\Big)$, and consequently*

$$C_n \sim \binom{\pi(n)+1}{2} \sim \frac{\pi(n)^2}{2} \sim \frac{n^2}{2\log^2 n}.$$

In particular the bound of Theorem 3.4(3) of [9] *is asymptotically an equality.*

Proof. By Corollary 7 the difference equals

$$C_n - \binom{\pi(n)+1}{2} = \sum_{x \leq n \text{ composite}} \big(\pi(\operatorname{lpf} x) - 1\big),$$

a sum of non-negative terms. If $x \leq n$ is composite then $\operatorname{lpf} x \leq \sqrt{x} \leq \sqrt{n}$, so each term is at most $\pi(\sqrt{n}) - 1$, and there are fewer than n composites $\leq n$; this gives the upper bound. The prime number theorem gives $\pi(\sqrt{n}) \sim$

$2\sqrt{n}/\log n$, whence

$$n\big(\pi(\sqrt{n})-1\big) = O\left(\frac{n^{3/2}}{\log n}\right),$$

while

$$\binom{r+1}{2} \sim \frac{\pi(n)^2}{2} \sim \frac{n^2}{2\log^2 n},$$

which dominates it. ∎

Corollary 19. *$C_n =$ A182843$(n+1)$.*

Proof. A182843(m) is defined as the number of composite integers $\geq m$ all of whose proper divisors are less than m. A composite W has largest proper divisor $W/\operatorname{lpf} W$, so the defining condition for $m = n+1$ reads $W > n$ and $W \leq n \operatorname{lpf} W$. These are exactly the W corresponding to minimal generators of I_n: a monomial of weight W lies in $G(I_n)$ precisely when $W > n$ and $W/p \leq n$ for every prime $p \mid W$. Of those last conditions only one needs checking, namely the one at $p = \operatorname{lpf} W$, because W/p is largest for the smallest p; so $W/p \leq n$ for all $p \mid W$ is equivalent to $W/\operatorname{lpf} W \leq n$, that is, to $W \leq n \operatorname{lpf} W$. ∎

The upper bound in Theorem 18 is far from best possible: almost all integers are even and contribute nothing. We now determine the error term exactly. Write

$$S(n) = C_n - \binom{\pi(n)+1}{2} = \sum_{x \leq n \text{ composite}} \big(\pi(\operatorname{lpf} x) - 1\big), \tag{19}$$

and let $S_2(n)$ be the part of the sum coming from x with $\Omega(x) = 2$.

Lemma 20. *$S_2(n) = \sum_{p \leq \sqrt{n}} \big(\pi(p)-1\big)\big(\pi(n/p) - \pi(p) + 1\big)$ (the sum over primes), and $0 \leq S(n) - S_2(n) \ll n^{4/3}/\log^2 n$.*

Proof. If $\Omega(x) = 2$ then $x = pq$ with $p \leq q$ prime, $\operatorname{lpf} x = p$ and $p \leq \sqrt{n}$; for fixed p the admissible q are the primes in $[p, n/p]$, of which there are $\pi(n/p) - \pi(p) + 1$, and each contributes the weight $\pi(p) - 1$. If instead $\Omega(x) \geq 3$ then $(\operatorname{lpf} x)^3 \leq x \leq n$, so $\operatorname{lpf} x \leq n^{1/3}$, and x is a multiple of $\operatorname{lpf} x$; hence

$$0 \leq S(n) - S_2(n) \leq \sum_{p \leq n^{1/3}} \big(\pi(p)-1\big)\frac{n}{p} \ll n \sum_{p \leq n^{1/3}} \frac{1}{\log p} \ll \frac{n^{4/3}}{\log^2 n},$$

by $\pi(p) \ll p/\log p$ and then $\sum_{p \leq Y} 1/\log p \ll Y/\log^2 Y$. ∎

Throughout what follows we abbreviate

$$X = \sqrt{n}, \qquad M = \log X = \tfrac{1}{2}\log n, \qquad P = \pi(\sqrt{n}), \tag{20}$$

and we write

$$g(t) = \big(\pi(t) - 1\big)\big(\pi(n/t) - \pi(t) + 1\big). \tag{21}$$

With this notation Lemma 20 reads

$$S_2(n) = \sum_{p \leq X} g(p), \tag{22}$$

the sum being over primes.

The next lemma is the reason the rest is short. It removes a full third of the asymptotics from the analysis, replacing it by an exact elementary sum.

Lemma 21 (Exact splitting)**.** *Let $p_1 < p_2 < \cdots < p_P$ be the primes $\le \sqrt{n}$, and put*

$$T(n) = \sum_{j=1}^{P} (j-1)\,\pi(n/p_j).$$

Then

$$S_2(n) = T(n) - \frac{(P-1)P(2P-1)}{6}. \tag{23}$$

Proof. Since p_j is the j-th prime, $\pi(p_j) = j$. So the summand of (22) at $p = p_j$ is

$$g(p_j) = (j-1)\big(\pi(n/p_j) - j + 1\big) = (j-1)\pi(n/p_j) - (j-1)^2,$$

and summing over j,

$$S_2(n) = T(n) - \sum_{j=1}^{P}(j-1)^2 = T(n) - \sum_{i=1}^{P-1} i^2 = T(n) - \frac{(P-1)P(2P-1)}{6}. \qquad \square$$

Remark 22. Lemma 21 explains where the constant $\frac{2}{3}$ comes from. Dividing (23) by P^3 and expanding the second term *exactly*,

$$\frac{S_2(n)}{P^3} = \frac{T(n)}{P^3} - \frac{1}{3} + \frac{1}{2P} - \frac{1}{6P^2}. \tag{24}$$

So $\frac{2}{3} = 1 - \frac{1}{3}$, where the 1 is the one thing still requiring the prime number theorem — that $T(n) \sim P^3$, Proposition 25 below — and the $\frac{1}{3}$ is $\sum i^2 \sim P^3/3$, which requires nothing at all.

The substitution governing T is $p = Xe^{-a}$, under which $n/p = Xe^{a}$; the parameter $a = \log(X/p) \ge 0$ measures how far below $\sqrt{n}$ the smaller prime factor lies. Note that the summand of T is, to leading order, *constant* in a: the two factors $\pi(p) - 1 \approx (X/M)e^{-a}$ and $\pi(n/p) \approx (X/M)e^{a}$ have reciprocal exponentials. This is why no Riemann sum is needed below — only a count of the primes involved.

Lemma 23. *Let A_0 be an absolute constant for which $|\pi(y) - y/\log y| \le A_0\, y/\log^2 y$ $(y \ge 2)$. There is an absolute constant A_1 such that for every $W \ge 1$, every n with $M \ge 2W$, and every real $a \in [0, W]$,*

$$\begin{aligned} \pi\big(Xe^{-a}\big) &= \frac{X}{M}\left(e^{-a} + \theta_1 \frac{A_1(W+1)}{M}\right), \\ \pi\big(Xe^{a}\big) &= \frac{X}{M}\left(e^{a} + \theta_2 \frac{A_1(W+1)e^{W}}{M}\right), \end{aligned} \tag{25}$$

where $|\theta_1|, |\theta_2| \le 1$.

The error constant in the second line grows with W, because e^a itself does; this is what limits how fast W may be allowed to grow with n.

Proof. For $y = Xe^{\mp a}$ we have $\log y = M \mp a$, and $0 \le a \le W \le M/2$ gives $\frac{1}{2}M \le \log y \le \frac{3}{2}M$; in particular $y \ge \sqrt{X} \ge M \ge 2$, so the hypothesis on A_0 applies. We also use $X \ge M^2$ freely below, which is automatic rather than an assumption: $X = e^M$, and e^M/M^2 attains its minimum $e^2/4 > 1$ at $M = 2$.

First line. By hypothesis on A_0,

$$\pi\big(Xe^{-a}\big) = \frac{Xe^{-a}}{M-a} + \theta\, A_0 \frac{Xe^{-a}}{(M-a)^2}, \qquad |\theta| \le 1,$$

and since $M - a \ge \frac{1}{2}M$ and $e^{-a} \le 1$ the second term is at most $4A_0X/M^2$. For the first, $0 \le a/M \le \frac{1}{2}$ gives

$$\frac{e^{-a}}{1-a/M} - e^{-a} = e^{-a} \cdot \frac{a/M}{1-a/M}, \qquad |\cdot| \le \frac{2a}{M} \le \frac{2W}{M},$$

so the first line holds with $A_1 \ge 2 + 4A_0$.

Second line. The same computation with $\log y = M + a$, now using the trivial $M + a \ge M$ together with $e^a \le e^W$, bounds the two error terms by A_0e^W/M and We^W/M, hence by $(W + A_0)e^W/M$ in total, after division by X/M. □

Lemma 24. *There is an absolute constant c_0 such that, for every W with $1 \le W \le M/2$,*

$$\sum_{p \le Xe^{-W}} \pi(p)\,\pi(n/p) \;\le\; c_0\, e^{-W}\, \frac{X^3}{M^3} \;+\; c_0\, \frac{X^3}{M^3} \cdot \frac{\log n}{n^{1/4}}.$$

The same bound holds for $\sum_{p \le Xe^{-W}} g(p)$ and for $\sum_{p \le Xe^{-W}} (\pi(p)-1)\pi(n/p)$, both summands being at most $\pi(p)\pi(n/p)$.

Proof. Let c_1 be an absolute constant with $\pi(y) \le c_1 y/\log y$ (Chebyshev), and write $L = \log n = 2M$. Split at $n^{1/4}$.

Range $p \le n^{1/4}$. Here $\log(n/p) \ge \frac{3}{4}L$, so

$$\pi(p)\,\pi(n/p) \;\le\; c_1^2 \frac{n}{\log p \cdot \log(n/p)} \;\le\; \frac{4c_1^2}{3\log 2} \cdot \frac{n}{L},$$

and there are at most $\pi(n^{1/4}) \le 4c_1 n^{1/4}/L$ such primes, giving a constant times $n^{5/4}/L^2$; relative to $X^3/M^3 = 8n^{3/2}/L^3$ that is a constant times $L/n^{1/4}$.

Range $n^{1/4} < p \le Xe^{-W}$. Here $\log p \ge \frac{1}{4}L$ and $\log(n/p) \ge \frac{1}{2}L$, so $\pi(p)\pi(n/p) \le 8c_1^2 n/L^2$, while

$$\pi\big(Xe^{-W}\big) \;\le\; c_1 \frac{Xe^{-W}}{M-W} \;\le\; 2c_1 \frac{Xe^{-W}}{M} = 4c_1 \frac{Xe^{-W}}{L}$$

by $W \le M/2$. The product is at most $32c_1^3 e^{-W} n^{3/2}/L^3 = 4c_1^3 e^{-W} X^3/M^3$. □

Proposition 25. *There are absolute constants C_0 and n_0 such that for $n \ge n_0$*

$$\left| \frac{M^3}{X^3}\, T(n) - 1 \right| \;\le\; C_0\, \frac{\log M}{\sqrt{M}}.$$

Proof. Fix $W \geq 1$ with $M \geq 2W$, to be chosen. Split T at Xe^{-W}; by Lemma 24 the part with $p \leq Xe^{-W}$ is at most $c_0(e^{-W} + \log n/n^{1/4})X^3/M^3$.

In the main range $Xe^{-W} < p \leq X$ put $a = \log(X/p) \in [0, W)$. By (25),

$$\big(\pi(p) - 1\big)\pi(n/p) = \frac{X^2}{M^2}\Big(e^{-a} + \theta_1 \tfrac{A_1(W+1)}{M} - \tfrac{M}{X}\Big) \times \Big(e^{a} + \theta_2 \tfrac{A_1(W+1)e^W}{M}\Big).$$

Expanding, the leading term is $e^{-a}e^{a} = 1$, and every other term carries a factor $A_1(W+1)e^W/M$ or M/X; using $e^{-a} \leq 1$, $e^a \leq e^W$ and $X \geq M^2$, all of them together are at most $A_2(W+1)e^W/M$ in absolute value, with A_2 absolute. Hence

$$\sum_{Xe^{-W} < p \leq X} \big(\pi(p) - 1\big)\pi(n/p) = \frac{X^2}{M^2}\Big(1 + \theta_3\, \tfrac{A_2(W+1)e^W}{M}\Big) \times \#\{\, p : Xe^{-W} < p \leq X \,\},$$

$|\theta_3| \leq 1$ — this is where the summand being constant to leading order does the work: no partition of $[0, W]$ and no Riemann sum is needed, only the number of primes in the range. That number is, by (25) again,

$$\pi(X) - \pi\big(Xe^{-W}\big) = \frac{X}{M}\Big(1 - e^{-W} + \theta_4\, \tfrac{2A_1(W+1)}{M}\Big).$$

Multiplying, and adding the tail bound,

$$\frac{M^3}{X^3}T(n) = 1 + O\!\left(e^{-W} + \frac{(W+1)e^W}{M} + \frac{\log n}{n^{1/4}}\right),$$

with an absolute implied constant. Now take $W = \frac{1}{2}\log M$, legitimate for $M \geq e^2$ since then $W \geq 1$ and $M \geq 2W$. It gives $e^{-W} = M^{-1/2}$ and $(W+1)e^W/M = (\frac{1}{2}\log M + 1)/\sqrt{M}$, both $O(\log M/\sqrt{M})$, while $\log n/n^{1/4}$ is smaller than either. □

Theorem 26. *There are absolute constants C and n_0 such that for all $n \geq n_0$*

$$\left| \frac{S_2(n)}{\pi(\sqrt{n})^3} - \frac{2}{3} \right| \;\leq\; C\,\frac{\log\log n}{\sqrt{\log n}}. \tag{26}$$

In particular, as $n \to \infty$,

$$S(n) \;\sim\; \tfrac{2}{3}\,\pi(\sqrt{n})^3 \;\sim\; \frac{16}{3}\cdot\frac{n^{3/2}}{\log^3 n},$$

and consequently

$$C_n = \binom{\pi(n)+1}{2} + \Big(\tfrac{16}{3} + o(1)\Big)\frac{n^{3/2}}{\log^3 n}.$$

In particular the upper bound of Theorem 18 overshoots by a factor $\asymp \log^2 n$.

Proof. By (24) it suffices to bound $T(n)/P^3 - 1$, since the remaining terms $1/(2P) - 1/(6P^2)$ are $O(1/P) = O(\log n/\sqrt{n})$, far smaller than the claimed bound. By Lemma 23 with $a = 0$ and $W = 1$, $P = \pi(X) = (X/M)(1 + 2\theta A_1/M)$, so

$$\frac{T(n)}{P^3} = \frac{M^3}{X^3}T(n)\cdot\Big(\frac{X}{MP}\Big)^3 = \Big(1+O\big(\tfrac{\log M}{\sqrt{M}}\big)\Big)\Big(1+O\big(\tfrac{1}{M}\big)\Big) = 1+O\Big(\frac{\log M}{\sqrt{M}}\Big)$$

by Proposition 25. Substituting $M = \frac{1}{2}\log n$ gives (26).

For the asymptotic statements, $P = \pi(\sqrt{n})$ and, by the prime number theorem, $\pi(\sqrt{n})^3 \sim (2\sqrt{n}/\log n)^3 = 8n^{3/2}/\log^3 n$, so $S_2(n) \sim \frac{2}{3}\pi(\sqrt{n})^3 \sim \frac{16}{3}n^{3/2}/\log^3 n$. Lemma 20 transfers this to S, and Corollary 7 to C_n. □

Remark 27. Five comments on the proof, the first two of which are traps.

(1) *No sieve theory is needed, and reaching for it gives the wrong answer.* The mass sits where p and n/p are both within a bounded factor of $\sqrt{n}$, that is, at $u = \log(n/p)/\log p \to 1$; and there Φ degenerates to the exact identity $\Phi(x, y) = \pi(x) - \pi(y) + 1$, valid for $y \geq \sqrt{x}$. The one-term Buchstab estimate $\Phi(x, y) \approx x\,\omega(u)/\log y$, with ω Buchstab's function, fed into (19), returns 8 rather than 16/3; and it is uniform only for $u \geq 2$ [19], which excludes precisely the region carrying the mass. Lemma 21 says exactly what goes missing. On $1 \leq u \leq 2$ one has $\omega(u) = 1/u$, so the one-term estimate amounts to replacing the second factor of g by $\pi(n/p)$ alone, that is, to dropping the $-\pi(p) + 1$. What is dropped is therefore $\sum_j (j-1)^2 = (P-1)P(2P-1)/6 \sim P^3/3$, which is $\frac{8}{3}n^{3/2}/\log^3 n$; restoring it gives $8 - \frac{8}{3} = \frac{16}{3}$. So the discrepancy is not a subtlety about Φ at all, but the exact square sum that Lemma 21 peels off.

(2) *The convergence is slow and not monotone*, so a numerical check over too short a range is misleading. The raw quotient $S(n)\log^3 n/n^{3/2}$ peaks near 10.9 around $n = 7\cdot 10^4$ and is still 9.55 at $n = 10^7$, against a limit of $16/3 = 5.33\ldots$. Most of that discrepancy, about two thirds of it on a logarithmic scale over the range tabulated, is the difference between $\pi(\sqrt{n})$ and $2\sqrt{n}/\log n$, which is cubed here; the factor $\big(\pi(\sqrt{n})\log n/2\sqrt{n}\big)^3$ is $1 + 6/\log n$ to first order, and is 1.56 at $n = 10^6$. Measured instead against $\frac{2}{3}\pi(\sqrt{n})^3$, which absorbs that factor entirely, the picture is much better behaved, though still not monotone: see Figure 2.

(3) *The proof uses no Riemann sums, no partition of the range, and no interchange of limits*: W is an explicit function of n, and (26) is a single estimate. That is a consequence of Lemma 21, which removes the $\frac{1}{3}$ exactly, together with the observation that what remains has a summand constant to leading order.

(4) *The rate is not sharp.* It is limited by the truncation at $a = W$, not by the arithmetic input: at $n = 10^{14}$ the left-hand side of (26) is 0.011, against 0.61 for $\log\log n/\sqrt{\log n}$. How much better the truth is we do not claim; multiplying the computed deviations by $\log n$ gives $0.55, 0.92, 0.59, 0.68, 0.54, 0.47, 0.37$ at $n = 10^6, \ldots, 10^{14}$, which neither settles nor rules out a rate of $1/\log n$. Sharpening the theorem would mean treating the whole range $0 \leq a \leq M$ rather than discarding $a > W$.

(5) *The hypotheses can be weakened* if one wants only the asymptotic and no rate. Lemma 23 was stated with the prime number theorem in the sharp form $\pi(y) = y/\log y + O(y/\log^2 y)$ because that is what produces an explicit error. The bare prime number theorem $\pi(y) \sim y/\log y$ already suffices for $S(n) \sim \frac{2}{3}\pi(\sqrt{n})^3$: for a in a *fixed* range $[0, W]$ the arguments $Xe^{\pm a}$ tend to infinity with n, so (25) holds with the error terms replaced by an unspecified $\varepsilon_W(n) \to 0$, uniformly in a; one then lets $W \to \infty$ afterwards, the tail of Lemma 24 being $O(e^{-W})$. This matters to anyone wishing to formalise the argument in a proof assistant, since it removes the need for a prime number theorem *with* an error term.

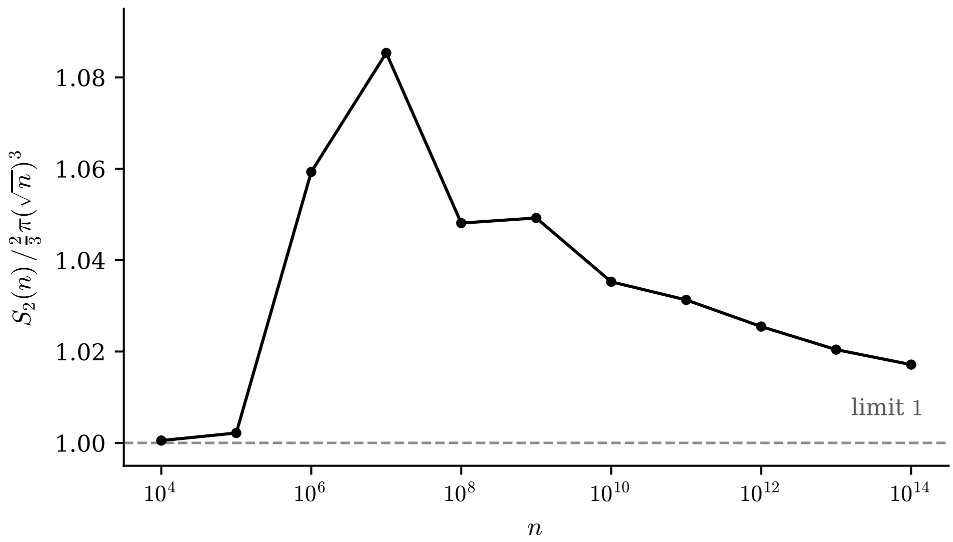

FIGURE 2. The quotient $S_2(n)/\frac{2}{3}\pi(\sqrt{n})^3$ against n, on a logarithmic n-axis, for $n = 10^4, \dots, 10^{14}$. Theorem 26 asserts that this tends to 1. It does, but only after rising to a maximum of 1.085 at $n = 10^7$, and with a further slight rise between 10^8 and 10^9; at $n = 10^{14}$ it is still 1.017. The two smallest values, at 10^4 and 10^5, sit within 0.3% of 1 by coincidence rather than for any reason, which is a useful warning on its own. Values for $n \leq 10^9$ are from a least-prime-factor sieve, those beyond from an $O(n^{3/4})$ prime-counting recursion [29].

7. THE SOCLE, AND THE TYPE

This section is routine: every ingredient is standard, and for I_n the evaluation is a one-line application. We include it because the answer is a quantity that has already appeared twice above, and because the contrast with the unitary case in Remark 31 seems worth recording.

Recall the standard multigraded reduction: for a monomial ideal $I \subsetneq S$, the socle $(0 :_{S/I} \mathfrak{m})$ is spanned by the residues of those monomials $u \notin I$

n	C_n	$\binom{\pi(n)+1}{2}$	ratio
10^2	362	325	1.1138
10^3	14 957	14 196	1.0536
10^4	769 090	755 835	1.0175
10^5	46 233 662	46 008 028	1.0049
10^6	3 084 943 710	3 081 007 251	1.0013
10^7	220 905 096 960	220 832 955 910	1.00033

Table 1. C_n against the lower bound of Theorem 3.4(3) of [9].

with $x_i u \in I$ for every i — because S/I is $\mathbb{Z}^r$-graded with one-dimensional components, so the annihilator is a multigraded subspace and hence spanned by the monomials in it. See [12] or [13].

Proposition 28. *Let $n \geq 2$. Then $\operatorname{soc}(\Gamma_n)$ is spanned by the monomials of weight m with $n/2 < m \leq n$; under the identification of the K-basis of Γ_n with $\{1, \dots, n\}$,*

$$\operatorname{soc}(\Gamma_n) = \bigoplus_{n/2<m\leq n} K \cdot m, \qquad \dim_K \operatorname{soc}(\Gamma_n) = \left\lceil \frac{n}{2} \right\rceil.$$

Proof. By the reduction above, u is a socle monomial exactly when $w(u) \leq n$ and $w(x_i u) = p_i w(u) > n$ for every $i = 1, \dots, r$. Since $p_1 = 2 \leq p_i$ for all i, the condition at $i = 1$ implies all the others, so the system reduces to that single binding constraint: $2w(u) > n$. So the socle monomials are those of weight in $(n/2, n]$, and there are $n - \lfloor n/2 \rfloor = \lceil n/2 \rceil$ of them. ☐

Corollary 29. *For $n \geq 2$ the Cohen–Macaulay type of Γ_n is*

$$\operatorname{type}(\Gamma_n) = \beta_r^S(\Gamma_n) = C_{n,1} = \Phi(n, 2) = \left\lceil \frac{n}{2} \right\rceil,$$

and Γ_n is Gorenstein if and only if $n \leq 2$.

Proof. For an Artinian module the last Betti number over S is the socle dimension, so the first two equalities are Proposition 28. For the third, Eliahou–Kervaire gives $\beta_i^S = \sum_v C_{n,v}\binom{r-v}{i-1}$, and at $i = r$ the binomial vanishes unless $v = 1$, leaving $\beta_r^S = C_{n,1}$; the last equality is Theorem 4. Gorenstein means type 1, i.e. $\lceil n/2 \rceil = 1$, i.e. $n \leq 2$. ☐

Remark 30. Corollary 29 is not an independent third sighting of $\lceil n/2 \rceil$, and it would be misleading to present it as one. For an Artinian Golod quotient the denominator of (17) is the list of Betti numbers,

$$D_n(t) = 1 - \sum_{i\geq 1} \beta_i^S \, t^{i+1};$$

since $\operatorname{pd}_S \Gamma_n = r$, its leading coefficient is $-\beta_r^S = -\operatorname{type}(\Gamma_n)$, and dividing by the monic $(1+t)^{r-\ell_1}$ preserves it. So clause (6) of Conjecture 4.6 *is* the assertion that the type is $\lceil n/2 \rceil$, and the proof of clause (6) given above — "the top-degree term arising only from $j = 1$" — is the Eliahou–Kervaire computation $\beta_r^S = C_{n,1}$ in generating-function clothing. What does remain is a genuine coincidence of two countings: the socle counts the integers in

$(n/2, n]$, while $\Phi(n, 2)$ counts the odd integers in $[1, n]$. Different sets, of the same size.

Remark 31. The reason Proposition 28 is a one-liner is that the socle condition is "$p_i w(u) > n$ for *every* prime $p_i \le n$", which collapses because the least prime is 2. The analogue for the *unitary* truncations does not collapse: there the condition reads "$kp > n$ for every prime p coprime to k", and [11, §7] obtains the socle dimension only asymptotically, as δn with

$$\delta = \frac{1}{2} + \sum_{i \ge 1} \frac{p_i^{-1} - p_{i+1}^{-1}}{p_1 p_2 \cdots p_i} \approx 0.6077.$$

For Dirichlet convolution nothing of the sort is needed: the socle is exactly the integers in $(n/2, n]$, for every n, so its density is $1/2$ in the limit and $1/2 + O(1/n)$ at every finite stage. The asymmetry between the two convolutions is the interesting content here, rather than the statement itself.

Remark 32. Grading by $\lambda = \Omega$, the algebra Γ_n is *level* — socle concentrated in one degree — exactly for $n \le 3$. Indeed for $n \ge 4$ let 2^k be the largest power of 2 with $2^k \le n$; then $2^k > n/2$, so 2^k is a socle element of degree $k \ge 2$, while Bertrand's postulate applied to $\lfloor n/2 \rfloor$ supplies a prime q with $n/2 < q \le n$, a socle element of degree 1.

8. Related work

Two remarks on the surrounding literature, both of which bear on how the result above should be read.

First, on the algebraic side, the ring Γ itself continues to be studied — recently in [21], which shows it is neither Noetherian nor Artinian, constructs several families of prime ideals, and computes its Krull dimension to be infinite. Its finite-dimensional truncations appear not to have been taken up by anyone else, but they were taken up once more by the present author, from a different direction: $\{1, 2, \ldots, n\}$ is a *shifted multicomplex*, the complement of a strongly stable ideal being to a multicomplex what a shifted simplicial complex is to a squarefree one, and [14] accordingly computes the spectrum of its Laplacian. By the theorem of Duval and Reiner [15] that spectrum is integral, and [14, Cor. 14] identifies its multiplicities as the arithmetic counts

$$t_{i,k}(n) = \#\{\, 1 < m \le n : \Omega(m) = k,\ p_i \mid \mathrm{sfp}(m) \,\},$$

sfp denoting the squarefree part. These are not the counts $C_{n,v}$ of the present note — they fix the total degree and impose a divisibility condition, rather than fixing the least support — so neither result subsumes the other, and it is a natural question, and the exact analogue of Theorem 4, whether $t_{i,k}$ too admits a sieve-theoretic closed form. Artinian truncations of Stanley–Reisner rings are an active topic [26], and Γ_n is an object of that general shape, but no arithmetic enters there.

Second, and more to the point, the bridge Theorem 4 builds appears to be new. The asymptotic behaviour of $\Phi(x, y)$ is classical: Buchstab [16] obtained $\Phi(x, x^{1/u}) \sim u\,\omega(u)\,x/\log x$ in terms of the function that now bears his name, and de Bruijn [17] gave an approximation uniform for all $x \ge y \ge 2$; see [27] for a textbook account, [19] for numerically explicit estimates,

[20] for short intervals, and [23] for the regime of fixed p, which is the one closest to ours. We have not been able to find anywhere in the literature a connection between these counts and any invariant of a free resolution; the two subjects appear not to cite each other at all. (A search for "Buchstab" in commutative algebra is apt to return instead the bigraded Betti numbers of face rings studied by V. M. Buchstaber, which is an unrelated subject and an unrelated person.)

9. What is next

Theorem 4 moves the combinatorics of $G(I_n)$ wholesale into sieve theory, and the questions it raises are correspondingly analytic.

Open problem 33. Give asymptotics for the whole profile $v \mapsto C_{n,v} = \Phi(n, p_v)$, *uniformly* in v, and deduce asymptotics for the graded Betti numbers of Γ_n through Corollary 4.2 of [9]. The behaviour of $\Phi(x, y)$ is classical [16, 17, 27], but here $y = p_v$ sweeps the whole interval $[2, n]$ as v does, so what is wanted is a statement uniform across all of it, including the two degenerate ends $v = 1$, where $\Phi(n, 2) = \lceil n/2 \rceil$, and $v = \pi(n)$, where $\Phi(n, p_r) = 1$. The uniform approximation of [17] and the explicit estimates of [19] are the natural starting points.

Open problem 34. Theorem 4 counts minimal generators by least support only. Is there a comparable sieve-theoretic description of the refined counts $C_{n,v,d}$, which additionally fix the total degree $d = \Omega$? A natural guess is a Φ-analogue restricted to integers with exactly d prime factors; the tables in Figure 2 of [9] are the data to test it against.

Open problem 35. Dirichlet convolution is the coarsest of Narkiewicz's regular convolutions. Carry out the analysis of [9] and of this note for the unitary, ternary, and general greedy convolutions: which truncation ideals are stable, and does an analogue of Theorem 4 hold? For the unitary case the truncations were treated by Stanley–Reisner methods in [10, 11], and it is not clear that a sifting function appears there at all. This is work in progress by the author, who intends to treat the unitary case next.

Open problem 36. The elementary half of this note involves no homological algebra and is within reach of a proof assistant. Part of it has been formalised in Lean 4 and Mathlib, and is available in [29]: Corollary 6, and the whole polynomial argument of Section 5 — Lemma 13, Proposition 15 and clauses (1), (2), (4), (5) of Theorem 16 — are proved outright, with no `sorry` and on the standard axioms. What remains is (i) clauses (3) and (6), which are degree and leading-coefficient computations, fiddly rather than deep; (ii) Propositions 10–11; and (iii) Theorems 3 and 4 themselves. Item (iii) is the real obstruction, and it is the same one that puts Section 5's homological input out of reach: the formalisation takes $C_{n,v} = \Phi(n, p_v)$ as a definition and checks it numerically, because $G(I_n)$ cannot even be stated without monomial ideals, which Mathlib does not have — any more than it has minimal free resolutions, graded Betti numbers or Golod rings.

Finally, three integer sequences arising here deserve to be in the OEIS. C_n is A182843, whose entry records none of the above; the sequence $C_{n,2}$ and the triangle $C_{n,v}$ appear not to be present at all.

Notation

Collected for reference; equation numbers point at the defining display.

Arithmetical	
$p_1 = 2 < p_2 < \cdots$	the primes
$\pi(x)$	the number of primes $\leq x$
$\Omega(m)$, also λ	number of prime factors of m, with multiplicity
$\operatorname{lpf}(m)$	least prime factor of $m > 1$
$\operatorname{sfp}(m)$	squarefree part of m
$\Phi(x, y)$	Legendre's sifting function, (8)
$R_p(y)$	number of p-rough integers $\leq y$
$\omega(u)$	Buchstab's function, *not* the count of distinct prime factors
Of this note and of [9]	
Γ, Γ_n	the ring of number-theoretic functions, and its truncation
S, I_n	$K[x_1, \ldots, x_r]$ with $r = \pi(n)$, and the truncation ideal
$w(m)$	weight of a monomial, $\prod_i p_i^{a_i}$
$G(I_n)$	the minimal monomial generating set
$\min(m)$, $\max(m)$	least and greatest index occurring in m
$\operatorname{supp}(m)$	the set of indices occurring in m
$C_{n,v}$, C_n	generator counts by least support, and their total, (7)
$E(n, v)$	composites $\leq n$ with every prime factor $> p_v$, (15)
D_n, q_n	denominators of the Poincaré–Betti series, (17)
ℓ_1, ℓ_2	the exponents of Conjecture 4.6
$S(n)$, $S_2(n)$	the error term (19), and its $\Omega = 2$ part
$T(n)$, P	Lemma 21; $P = \pi(\sqrt{n})$
X, M	$\sqrt{n}$ and $\log \sqrt{n}$, (20)
soc	socle, Section 7

Four collisions are worth flagging. Against standard usage: ω here is Buchstab's function, not the number of *distinct* prime factors, which this note never uses. Against the accompanying computational report [29]: $T(n)$ is the sum of Lemma 21, while that report uses the same letter for an unrelated quantity in its $\Omega \geq 3$ table. And two internal ones, both harmless but worth naming: S is the polynomial ring while $S(n)$ is the error term (19), and q is a generic prime in Sections 3–4 while q_n is a polynomial in Section 5.

Acknowledgements

The whole subject of this note, and of [9] before it, is owed to **Johan Andersson**. It was he who suggested studying the truncations Γ_n in the first place, and he who pointed out that the ideals I_n are monomial — and that this ought, therefore, to be my cup of tea. It was, and the observation has now paid out twice.

His hope at the time was specific, and it is worth recording because a quarter of a century has now furnished a partial answer to it. Monomial ideals carry a great deal of theory; the truncations Γ_n are monomial and arithmetic at once; so, he reasoned, the algebraic machinery ought to run backwards and deliver number-theoretic conclusions for nothing. That is not what has happened. The traffic has gone almost entirely the other way. It is the algebra that has supplied the questions — Conjecture 4.6, which is in

the end the assertion that $v \mapsto C_{n,v}$ is exactly linear beyond $v = \ell_1(n)$; the error term in Theorem 18, which is a Mertens-type sum — and answering them as number theory is what has illuminated the algebra. Theorem 26 is the clearest case: a statement about the minimal free resolution of Γ_n, settled by an argument containing no algebra at all.

There is, I think, a structural reason for the one-way traffic, and stating it is more useful than lamenting it. Most of what commutative algebra knows about a monomial ideal is *extremal*: Macaulay's theorem, Kruskal–Katona, the Bigatti–Hulett–Pardue bound. Each of these constrains an arbitrary object with a given Hilbert function, and is therefore only as strong as the worst such object. But Γ_n is nowhere near extremal, and the resulting bounds are correspondingly slack. Concretely: $\{1, 2, \ldots, n\}$ is a multicomplex, so by Macaulay's theorem the sequence

$$d \;\longmapsto\; \#\{\, m \le n : \Omega(m) = d \,\},$$

the Landau counts, must be an M-sequence, which yields growth bounds free of any analytic input. They are true and they are useless. At $n = 200$ there are 46 primes, and Macaulay permits as many as 1081 integers with $\Omega = 2$; the true number is 62. No inequality of that kind is going to tell an analytic number theorist something he does not already know.

Where the transfer does work is where the algebra is *exact* rather than extremal. Eliahou–Kervaire gives the graded Betti numbers on the nose, and it is precisely that exactness which makes this note possible at all: the identity $C_{n,v} = \Phi(n, p_v)$ is a statement about an invariant that the algebra computes rather than bounds. Section 7 is a small instance of the same phenomenon, and [14] — where Duval–Reiner integrality forces the Laplacian spectrum of Γ_n to be a list of arithmetic counting functions — is another. So Johan's hope was not wrong so much as mis-aimed. The fruit is there; it is just not low-hanging, and it does not grow on the general-inequality branch of the tree.

* * *

Just as Geheimrat Goethe had the diligent Friedrich Wilhelm Riemer to help him write *Die Wahlverwandtschaften*, just as Grothendieck had Jean Dieudonné to help him complete EGA, I have **Claude**! What exactly that amounted to is set out in the disclosure section below.

* * *

It is a pleasure to acknowledge the computer algebra systems without which none of this would exist. The computations behind [9] were carried out in Maple: the generator counts of its Figures 1 and 2, and — assembled from those counts through the formulas of its Corollaries 4.2 and 4.4 — the Poincaré–Betti series of its Figure 3. In the companion papers on unitary convolution the division of labour was quite different. For [10, 11], Maple was used to *generate* Macaulay2 [24] input — procedures emitting ring and ideal declarations line by line — and Macaulay2 then computed the free resolutions, Betti numbers and Poincaré–Betti series. The homology of the relevant simplicial complex was computed with the Simplicial Homology package [18] of Dumas, Heckenbach, Saunders and Welker for GAP [4]; having found it torsion-free, the author wrote a short GAP programme to test

lex-shellability, which it duly was. Every computation in the present note was done in SageMath [8], and deliberately along a route different from the one taken in 1999 — the minimal generators of I_n are enumerated as integers rather than counted by the formulas under test — so that agreement between the two is evidence rather than tautology.

Disclosure of AI assistance

This manuscript was produced with substantial assistance from a large language model, Claude (Anthropic), and the extent of it is set out here rather than left to be inferred.

The errata of Section 4 were found by Claude while re-reading [9] with the author, by recomputing every table in that paper along a route independent of the one used there. Found likewise by Claude, working with the author in interactive sessions: Theorem 4 and its proof; the reduction of Conjecture 4.6 to the order of vanishing of D_n at $t = -1$ (Lemma 13 and Proposition 15); the asymptotic Theorem 18; the identification of C_n with A182843; Theorem 26 together with Lemmas 20, 23 and 24, and the two cautions now folded into Remark 27; and the contents of Section 7. Claude also located and verified the references, drew Figures 1 and 2, wrote the accompanying SageMath code and the Lean formalisation mentioned in Open Problem 36, and drafted this manuscript, with the author directing the exposition throughout.

Every numerical claim in the note was checked along a route independent of the formula being tested, and the reference list has been checked against primary sources. The author has verified the results to the best of his ability, made the final edits, and assumes full responsibility for any errors, mistakes or gaps that remain.

Funding

This research received no external funding. It was carried out as part of the author's regular duties at Linköping University, within the fraction of that employment allocated to research.

Conflicts of interest

The author declares no financial or non-financial conflicts of interest. The nature and extent of large-language-model assistance in producing this manuscript is disclosed in full in the section above.

References

[1] Annetta Aramova and Jürgen Herzog, *Koszul cycles and Eliahou–Kervaire type resolutions*, J. Algebra **181** (1996), no. 2, 347–370.
[2] E. D. Cashwell and C. J. Everett, *The ring of number-theoretic functions*, Pacific J. Math. **9** (1959), 975–985.
[3] Shalom Eliahou and Michel Kervaire, *Minimal resolutions of some monomial ideals*, J. Algebra **129** (1990), no. 1, 1–25.
[4] The GAP Group, *GAP — Groups, Algorithms, and Programming*, `https://www.gap-system.org`.
[5] E. S. Golod, *On the homology of some local rings*, Soviet Math. Dokl. **3** (1962), 745–749.

[6] Tor H. Gulliksen and Gerson Levin, *Homology of local rings*, Queen's Papers in Pure and Applied Mathematics, vol. 20, Queen's University, Kingston, Ont., 1969.
[7] Irena Peeva, *0-Borel fixed ideals*, J. Algebra **184** (1996), no. 3, 945–984.
[8] The Sage Developers, *SageMath, the Sage Mathematics Software System*, `https://www.sagemath.org`.
[9] Jan Snellman, *Truncations of the ring of number-theoretic functions*, Homology Homotopy Appl. **2** (2000), 17–27; arXiv:math/9904143.
[10] Jan Snellman, *The ring of arithmetical functions with unitary convolution: general truncations*, Int. J. Math. Game Theory Algebra **13** (2003), no. 6, 485–519; arXiv:math/0205242.
[11] Jan Snellman, *The ring of arithmetical functions with unitary convolution: the $[n]$-truncation*, Int. J. Math. Game Theory Algebra **13** (2003), no. 6, 485–519; arXiv:math/0208183.
[12] Jürgen Herzog and Takayuki Hibi, *Monomial Ideals*, Grad. Texts in Math., vol. 260, Springer, 2011.
[13] Ezra Miller and Bernd Sturmfels, *Combinatorial Commutative Algebra*, Grad. Texts in Math., vol. 227, Springer, 2005.
[14] Jan Snellman, *Laplacians on shifted multicomplexes*, arXiv:math/0606104.
[15] Art M. Duval and Victor Reiner, *Shifted simplicial complexes are Laplacian integral*, Trans. Amer. Math. Soc. **354** (2002), no. 11, 4313–4344.
[16] A. A. Buchstab, *Asymptotic estimates of a general number-theoretic function*, Mat. Sb. **44** (1937), 1239–1246. (Russian.)
[17] N. G. de Bruijn, *On the number of uncancelled elements in the sieve of Eratosthenes*, Indag. Math. **12** (1950), 247–256.
[18] Jean-Guillaume Dumas, Frank Heckenbach, B. David Saunders and Volkmar Welker, *Simplicial Homology*, a GAP package.
[19] Steve Fan, *Numerically explicit estimates for the distribution of rough numbers*, arXiv:2306.03347.
[20] Ofir Gorodetsky, *The variance of integers without small prime factors in short intervals*, Math. Z. **308** (2024), no. 4, Paper No. 59; arXiv:2111.00853.
[21] Amit Goswami, Steven Kleyn and Lauren Porrill, *On structures of the ring of arithmetical functions: prime ideals and beyond*, arXiv:2410.10824.
[22] Jürgen Herzog, Victor Reiner and Volkmar Welker, *Componentwise linear ideals and Golod rings*, Michigan Math. J. **46** (1999), no. 2, 211–223.
[23] Fred B. Holt, *On the counts of p-rough numbers*, arXiv:2308.07570.
[24] Daniel R. Grayson and Michael E. Stillman, *Macaulay2, a software system for research in algebraic geometry*, `https://macaulay2.com`.
[25] Jason McCullough and Irena Peeva, *Infinite graded free resolutions*, in: Commutative Algebra and Noncommutative Algebraic Geometry, Vol. I, Math. Sci. Res. Inst. Publ. **67**, Cambridge Univ. Press, 2015, pp. 215–257.
[26] Sora Miyashita, *Canonical traces of Artinian truncations of Stanley–Reisner rings*, arXiv:2608.15955.
[27] Gérald Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Grad. Stud. Math. **163**, Amer. Math. Soc., 2015.
[28] The On-Line Encyclopedia of Integer Sequences, published electronically at `https://oeis.org`, sequence A182843.
[29] Supporting code, raw computational output and working notes, `projects/dirichlet-truncations/` in the repository `https://gitlab.liu.se/jansn19/ai-workspace`.